ICT232 — MACHINE LEARNING FOUNDATIONS

FIRST OPEN ENDED

Duration: 2 Hours

Total Marks: 100

Instructions to Students

1. Answer **all questions in Section A**.
2. Answer **any three questions in Section B**.
3. Show clear reasoning in all answers.
4. Where examples are required, use practical workplace situations.
5. Marks are awarded for **explanation, judgement, and critical thinking**, not definitions alone.
6. Use of Generative AI is not allowed in the text

SECTION A —(40 Marks)

Question 1 (10 Marks)

A university wants to predict whether first-year students are likely to complete their studies successfully.

Available data:

- age
- attendance
- internet access
- previous grades

Required:

a) Explain why this is a machine learning problem rather than a traditional programming problem. (4 marks)

b) State one reason why fixed programming rules may fail in this situation. (3 marks)

c) Explain why historical student outcomes are important in building such a system. (3 marks)

Question 2 (10 Marks)

A hospital wants to analyse patient data using:

- age
- blood pressure
- sugar level
- diagnosis result

Required:

a) Identify the features and the target variable. (4 marks)

b) Explain why this is supervised learning. (3 marks)

c) State one practical risk if diagnosis labels contain errors. (3 marks)

Question 3 (10 Marks)

A bank wants to predict loan default.

Possible outputs are:

- default / no default
- expected financial loss

Required:

a) Which output requires classification? Explain. (4 marks)

b) Which output requires regression? Explain. (4 marks)

c) State one limitation of classification in banking decisions. (2 marks)

Question 4 (10 Marks)

A retail company has customer data containing:

- monthly spending
- number of visits
- transaction frequency

No target variable exists.

Required:

a) Explain why this is not supervised learning. (4 marks)

b) Recommend one suitable machine learning approach. (3 marks)

c) Explain one business benefit of using clustering in this situation. (3 marks)

SECTION B —(60 Marks)

Question 5 (20 Marks)

Machine learning changes the way intelligent systems are designed.

Required:

a) Distinguish clearly between traditional programming and machine learning using one practical example from finance, education, or healthcare. (10 marks)

b) Explain why machine learning is more suitable when rules are difficult to define manually. (10 marks)

Question 6 (20 Marks)

A company wants to predict employee attrition using:

- age
- salary
- years employed
- left company

Required:

a) Explain why this is supervised learning. (5 marks)

b) Identify the target variable. (3 marks)

c) Explain whether classification or regression is more appropriate and justify your answer. (7 marks)

d) State two important variables missing from this dataset that may improve prediction quality. (5 marks)

Question 7 (20 Marks)

Unsupervised learning helps organisations discover hidden patterns.

Required:

a) Explain what unsupervised learning means. (5 marks)

b) Distinguish clustering from classification. (7 marks)

c) Design one workplace problem where clustering is more useful than supervised learning. (8 marks)

Question 8 (20 Marks)

A student says:

“If a dataset contains numbers, regression should always be used.”

Required:

a) Explain why this statement is incorrect. (8 marks)

b) Give one example where numbers appear in features but classification remains correct. (6 marks)

c) Explain one mistake that may occur when students choose methods based only on variable appearance. (6 marks)

THE END